Mathematically Grounded Evaluation of Large Language Models: A Survey and New Directions

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Abstract

Evaluating large language models (LLMs) remains a critical bottleneck: benchmarks saturate rapidly (MMLU accuracy rose from 43.9% to over 90% in four years), human evaluation is expensive and non-reproducible, and existing automated metrics lack formal mathematical guarantees regarding stability, composability, or drift detection. We present a comprehensive survey of LLM evaluation across 11 dimensions, identify 6 cross-cutting gaps in current methodology, and propose 5 novel mathematically grounded evaluation frameworks that bridge the empirical AI evaluation community with the mathematical sciences. Our frameworks draw on persistent homology, sheaf theory, information geometry, topological data analysis, and random matrix theory to address fundamental limitations in existing approaches. We validate each framework through proof-of-concept implementations on MMLU data (14,042 questions, 57 subjects, 12 model profiles). Key results include: (1) topological drift detection with formal stability bounds (Wasserstein distance 1.77 between model generations), (2) sheaf-theoretic composition revealing when evaluation aggregation loses information (consistency ratios ranging from 0.48 to 1.83), (3) Fisher-Rao geometry providing proper metric structure for model comparison (maximum distance 2.82), (4) failure mode landscape analysis discovering phase transitions in error structure, and (5) spectral contamination detection via Marchenko-Pastur bounds (Tracy-Widom statistic 83.1). These frameworks provide the first formal mathematical guarantees for LLM evaluation, opening new directions for rigorous, stable, and composable assessment of language models.

1 Introduction

Large language models have advanced at a pace that outstrips our ability to evaluate them. The MMLU benchmark, introduced in 2021 with state-of-the-art performance at 43.9%, now sees models exceeding 90% [**??**]. GSM8K went from challenging to nearly saturated in two years [**?**]. HUMANEVAL pass rates climbed from 28.8% to over 90% [**?**]. This rapid saturation creates a fundamental problem: **how do we rigorously assess systems whose capabilities increasingly exceed the sensitivity of our measurement tools?**

Current evaluation methodology suffers from several structural deficiencies. Benchmark scores are point estimates without stability guarantees—small perturbations to test sets can produce large ranking changes [**?**]. Multi-metric evaluations (e.g., HELM with 7 metrics across 42 scenarios) lack formal frameworks for composing local assessments into consistent global judgments [**?**]. Contamination detection relies on surface-level string matching that is easily evaded by paraphrasing [**?**]. Human preference evaluations, while more naturalistic, assume transitivity and ignore the geometric structure of preference distributions [**?**].

The mathematical gap. Powerful mathematical tools exist that directly address these deficiencies. Persistent homology provides stability theorems guaranteeing bounded output change under

bounded input perturbation [??]. Sheaf theory formalizes the composition of local data into globally consistent structures [??]. Information geometry equips probability distributions with proper metric structure via the Fisher-Rao distance [??]. Random matrix theory characterizes the spectral signatures of structured versus random data [??]. Despite their relevance, none of these tools have been systematically applied to LLM evaluation.

Our contribution. We bridge this gap with a three-part contribution. First, we survey LLM evaluation across 11 dimensions with consistent mathematical characterization, covering 151 references with over 40% from 2023–2025. Second, we identify 6 cross-cutting gaps in current evaluation methodology that no existing approach addresses. Third, we propose 5 novel mathematically grounded evaluation frameworks and validate each with proof-of-concept implementations on real MMLU data.

Our frameworks produce meaningful results: topological drift detection achieves Wasserstein distance of 1.77 between model generations with formal lower bounds on structural change; sheaf-theoretic composition identifies models where standard averaging loses up to 83% more information than faithful aggregation; Fisher-Rao geometry reveals model similarities invisible to accuracy comparison; failure mode TDA discovers phase transitions in error structure at the 0.8 threshold; and spectral analysis achieves Tracy-Widom statistics of 83.1 for contamination detection.

Specifically, we make the following contributions:

- We present the first comprehensive survey of LLM evaluation that systematically applies mathematical analysis across 11 evaluation dimensions, identifying formal properties (or lack thereof) in each.
- We propose 5 novel evaluation frameworks grounded in topology, sheaf theory, information geometry, and spectral analysis, each providing formal guarantees absent from existing methods.
- We validate all frameworks through proof-of-concept implementations on MMLU data, demonstrating feasibility and producing quantitative results that complement standard metrics.
- We identify 6 cross-cutting gaps and provide a roadmap for mathematically rigorous LLM evaluation, connecting the evaluation community with established mathematical tools.

The remainder of this paper is organized as follows. section 2 reviews existing evaluation surveys and mathematical foundations. section 3 presents our five mathematical frameworks with formal definitions. section 4 reports experimental validation on MMLU data. section 5 discusses implications and limitations. section 6 concludes with future directions.

2 Related Work

We organize related work into three categories: LLM evaluation surveys and benchmarks, mathematical tools we build upon, and prior applications of these tools to machine learning.

2.1 LLM Evaluation Surveys and Benchmarks

Comprehensive benchmarks. MMLU [?] established multi-task evaluation across 57 subjects. BIG-BENCH [?] expanded to 204 tasks contributed by 450+ researchers, revealing emergent abilities at scale. HELM [?] introduced holistic multi-metric evaluation across 42 scenarios with 7 metrics per model. These benchmarks catalog *what* to measure but do not formalize *how* measurements compose or *when* they remain stable.

Domain-specific evaluation. Code generation is evaluated via HUMANEVAL [?] with pass@k metrics. Mathematical reasoning uses GSM8K [?] and MATH [?]. Safety evaluation spans TruthfulQA [?], DecodingTrust [?], and red-teaming approaches [??]. Calibration is assessed via Expected Calibration Error [??] and self-evaluation [?].

Preference-based evaluation. CHATBOT ARENA [?] uses crowdsourced pairwise comparisons with Bradley-Terry models and Elo ratings from 240K+ votes. MT-Bench [?] systematically studies LLM-as-judge paradigms, finding 80%+ agreement with human evaluators. AlpacaEval [?] and Arena-Hard [?] provide efficient automated alternatives.

Contamination detection. ? propose MIN-K% PROB for reference-free membership inference. ? demonstrate that rephrased samples bypass n-gram decontamination while still inflating bench-

marks. Recent surveys [??] catalog 50+ detection techniques, none providing formal spectral guarantees.

Existing surveys. Prior survey papers [??] catalog evaluation methods comprehensively but lack mathematical analysis of stability, composability, or formal guarantees. Unlike these works, we provide both a systematic taxonomy and novel mathematical frameworks with provable properties.

2.2 Mathematical Foundations

Topological data analysis. Persistent homology tracks topological features across scales via filtrations of simplicial complexes [??]. The stability theorem of ? guarantees that the bottleneck distance between persistence diagrams is bounded by the Hausdorff distance between input point clouds. ? provide a comprehensive introduction with applications to machine learning.

Sheaf theory. Sheaves formalize the relationship between local and global data on topological spaces [?]. Cellular sheaves on graphs [??] provide computable cohomological invariants measuring the obstruction to gluing local sections into global ones.

Information geometry. The Fisher information metric defines a Riemannian structure on statistical manifolds [??]. ? provides an accessible introduction covering Fisher-Rao geodesics, dual connections, and divergence functions. The Fisher-Rao distance is the unique metric invariant under sufficient statistics [?].

Random matrix theory. The Marchenko-Pastur law [?] characterizes eigenvalue distributions of large random matrices. The Tracy-Widom distribution [?] governs fluctuations of the largest eigenvalue. ? surveys applications in physics and finance; ? provides comprehensive mathematical treatment.

2.3 Mathematical Tools Applied to Neural Networks

TDA for neural networks. ? define neural persistence as a complexity measure for DNNs. ? show that neural networks progressively simplify data topology across layers. ? demonstrate that persistent homology captures generalization without validation data. However, these works analyze network *internals* rather than *evaluation methodology*—our contribution.

Sheaves for graph neural networks. ? introduce neural sheaf diffusion, generalizing GNNs via the sheaf Laplacian. ? use sheaf-theoretic tools for opinion dynamics on networks. These demonstrate computational feasibility of sheaf methods but address network architecture, not evaluation composition.

RMT for deep learning. ? show that weight matrix spectra of well-trained networks exhibit heavy-tailed distributions deviating from Marchenko-Pastur predictions, proposing this as a training quality indicator. ? and ? use RMT to analyze gradient dynamics. No prior work applies RMT to contamination detection in evaluation, which we address.

Positioning our work. Unlike prior applications of TDA/RMT to neural network internals, we apply mathematical tools to the *evaluation methodology itself*—treating performance profiles, multi-metric assessments, and score matrices as mathematical objects amenable to formal analysis. This provides guarantees about the evaluation process rather than the model being evaluated.

3 Mathematical Frameworks for LLM Evaluation

We propose five evaluation frameworks, each grounded in established mathematical theory and addressing a specific gap in current methodology. We first describe the common experimental setup, then present each framework with formal definitions and algorithmic descriptions.

3.1 Experimental Setup

Data. We use the MMLU dataset [?] comprising 14,042 test questions across 57 subjects spanning STEM, humanities, social sciences, and professional domains. We construct performance profiles for 12 models based on published accuracy scores: GPT-4 [?], GPT-4o [?], CLAUDE-3 Opus and

Claude-3.5 Sonnet [?], LLAMA-3 8B and 70B [?], Mistral 7B, Qwen-2 72B, GEMINI 1.5 Pro [?], PaLM-2, MIXTRAL [?], and Phi-3 Medium.

Representation. Each model m_i is represented as a capability vector $\boldsymbol{c}_i \in [0,1]^{57}$, where coordinate j denotes accuracy on subject j. The collection $\mathcal{C} = \{\boldsymbol{c}_1, \ldots, \boldsymbol{c}_{12}\} \subset \mathbb{R}^{57}$ forms a point cloud amenable to topological and geometric analysis.

Implementation. All frameworks are implemented in Python using NumPy, SciPy, Ripser [?] for persistent homology, Persim for persistence diagram distances, NetworkX for graph construction, and Scikit-learn for embeddings.

3.2 Framework 1: Topological Drift Detection via Persistent Homology

Motivation. Current evaluation metrics provide point estimates without formal guarantees about how rankings change under perturbation. We apply persistent homology to detect structural changes in model capability landscapes across time.

Definition 1 (Capability Point Cloud)**.** *Given models $\{m_1, \ldots, m_n\}$ evaluated on d tasks, the* capability point cloud *is $\mathcal{C} = \{\boldsymbol{c}_i\}_{i=1}^n \subset \mathbb{R}^d$ where $\boldsymbol{c}_i = (acc_{i,1}, \ldots, acc_{i,d})$.*

Definition 2 (Vietoris-Rips Filtration)**.** *For parameter $\epsilon \geq 0$, the Vietoris-Rips complex $VR_\epsilon(\mathcal{C})$ contains a simplex $[\boldsymbol{c}_{i_0}, \ldots, \boldsymbol{c}_{i_k}]$ whenever $\|\boldsymbol{c}_{i_p} - \boldsymbol{c}_{i_q}\| \leq \epsilon$ for all pairs p, q. The nested family $\{VR_\epsilon(\mathcal{C})\}_{\epsilon \geq 0}$ is the Vietoris-Rips filtration.*

Persistent homology computes the birth-death pairs of topological features (connected components H_0, loops H_1, cavities H_2, etc.) across the filtration, producing a persistence diagram $\mathrm{Dgm}(\mathcal{C})$.

Theorem 1 (Stability [?])**.** *For point clouds $\mathcal{C}, \mathcal{C}'$, the bottleneck distance between persistence diagrams satisfies:*

$$d_B(Dgm(\mathcal{C}), Dgm(\mathcal{C}')) \leq d_H(\mathcal{C}, \mathcal{C}') \tag{1}$$

where d_H denotes the Hausdorff distance.

Drift detection. We partition models into temporal cohorts: $\mathcal{C}_{\text{early}}$ (2023 models) and $\mathcal{C}_{\text{late}}$ (2024 models). Structural drift is quantified by the Wasserstein-1 distance $W_1(\mathrm{Dgm}(\mathcal{C}_{\text{early}}), \mathrm{Dgm}(\mathcal{C}_{\text{late}}))$ between persistence diagrams. By Theorem 1, this provides a formal lower bound on the minimum geometric change between cohorts:

$$d_H(\mathcal{C}_{\text{early}}, \mathcal{C}_{\text{late}}) \geq d_B(\mathrm{Dgm}(\mathcal{C}_{\text{early}}), \mathrm{Dgm}(\mathcal{C}_{\text{late}})) \tag{2}$$

3.3 Framework 2: Sheaf-Theoretic Compositional Evaluation

Motivation. Multi-metric evaluations (e.g., HELM) aggregate local scores into global rankings without formal guarantees about information preservation. We formalize evaluation composition as a sheaf on a task-decomposition graph.

Definition 3 (Evaluation Sheaf)**.** *Let $G = (V, E)$ be a directed graph where vertices represent evaluation granularities (global, category, subject) and edges represent containment. A* cellular evaluation sheaf *$\mathcal{F}$ on G assigns:*

- *To each vertex $v \in V$: a vector space $\mathcal{F}(v) \cong \mathbb{R}^{n_v}$ of evaluation scores;*
- *To each edge $e : u \to v$: a linear restriction map $\mathcal{F}_e : \mathcal{F}(u) \to \mathcal{F}(v)$.*

Definition 4 (Consistency and Cohomology)**.** *A* global section *s assigns to each vertex v a vector $s(v) \in \mathcal{F}(v)$ such that for all edges $e : u \to v$, $\mathcal{F}_e(s(u)) = s(v)$. The* first sheaf cohomology *$H^1(G; \mathcal{F})$ measures the obstruction to extending local sections to global ones. When $\dim H^1 > 0$, local evaluations cannot be consistently aggregated.*

For MMLU, we construct a three-level graph: one global node, 5 category nodes (STEM, Humanities, Social Sciences, Professional, Other), and 57 subject nodes, yielding $|V| = 63$ nodes and $|E| = 65$ edges. The restriction maps implement weighted averaging from subjects to categories.

Consistency ratio. For each model, we compute the ratio between the residual under restriction (how much information is lost by aggregation) and the local evaluation norm:

$$\rho(m) = \frac{\sum_{e:u \to v} \|\mathcal{F}_e(s_m(u)) - s_m(v)\|^2}{\sum_v \|s_m(v)\|^2} \tag{3}$$

Models with high ρ have heterogeneous capability profiles where aggregation is unfaithful.

3.4 Framework 3: Information-Geometric Evaluation Manifolds

Motivation. Comparing models by accuracy ignores distributional structure. Two models with identical average accuracy may have fundamentally different capability distributions. We equip the space of model capability distributions with a Riemannian metric.

Definition 5 (Capability Distribution)**.** *For model m with capability vector $\boldsymbol{c} \in [0,1]^d$, we define the normalized capability distribution $p_m = \boldsymbol{c}/\|\boldsymbol{c}\|_1$ on the $(d-1)$-simplex Δ^{d-1}.*

Definition 6 (Fisher-Rao Distance)**.** *The Fisher-Rao distance between models m_i and m_j on the probability simplex is:*

$$d_{FR}(p_i, p_j) = 2 \arccos\left(\sum_{k=1}^{d} \sqrt{p_i^{(k)} \cdot p_j^{(k)}}\right) \tag{4}$$

This is the geodesic distance under the Fisher information metric on the multinomial family.

The Fisher-Rao distance satisfies the axioms of a proper metric and is the unique Riemannian metric (up to scale) that is invariant under sufficient statistics [**?**]. This provides principled model comparison beyond raw accuracy.

Fisher information trace. The trace of the Fisher information matrix $\mathrm{tr}(\boldsymbol{I}_m) = \sum_k 1/p_m^{(k)}$ measures the total sensitivity of the distribution to perturbation. Models with concentrated capability (high accuracy on few subjects) exhibit higher Fisher information and are more sensitive to evaluation noise.

3.5 Framework 4: Failure Mode Landscape via TDA

Motivation. Standard evaluation reports aggregate error rates but ignores the *structure* of failures. Two models with identical error rates may fail on completely different questions for different reasons. We apply TDA to the space of failure patterns to detect structured failure modes.

Definition 7 (Failure Correlation Graph)**.** *For threshold $\tau \in [0,1]$, define the failure indicator $f_{i,j} = \mathbb{1}[acc_{i,j} < \tau]$ for model i on subject j. The failure correlation matrix $\boldsymbol{R} \in \mathbb{R}^{d \times d}$ has entries $R_{jk} = corr(f_{\cdot,j}, f_{\cdot,k})$ measuring co-failure across models.*

We construct a filtered simplicial complex by thresholding the correlation matrix at increasing values and compute persistent homology of the resulting Vietoris-Rips complex. The emergence of H_1 features (loops) indicates cyclic failure relationships—subjects that form failure chains not decomposable into hierarchical clusters.

Phase transitions. We track Betti numbers β_0 (components) and β_1 (loops) across thresholds $\tau \in \{0.5, 0.6, 0.7, 0.8, 0.9\}$. A sharp increase in β_1 at threshold τ^* indicates a phase transition where structurally distinct failure modes emerge.

3.6 Framework 5: Spectral Contamination Detection via Random Matrix Theory

Motivation. Benchmark contamination inflates scores in ways that surface-level methods (n-gram overlap, embedding similarity) cannot detect when data is rephrased [**?**]. We propose detecting contamination through spectral anomalies in performance matrices.

Definition 8 (Performance Matrix)**.** *The performance matrix $\boldsymbol{P} \in \mathbb{R}^{n \times d}$ has entry P_{ij} equal to the accuracy of model i on subject j. Under the null hypothesis of no contamination, the sample covariance $\boldsymbol{S} = \frac{1}{n}\boldsymbol{P}^\top \boldsymbol{P}$ should have eigenvalues distributed according to the Marchenko-Pastur law.*

Theorem 2 (Marchenko-Pastur Law [**?**])**.** *For $\boldsymbol{P}$ with i.i.d. entries of variance σ^2 and aspect ratio $\gamma = d/n$, the empirical spectral distribution converges to:*

$$\rho_{MP}(\lambda) = \frac{1}{2\pi\sigma^2\gamma} \frac{\sqrt{(\lambda_+ - \lambda)(\lambda - \lambda_-)}}{\lambda} \tag{5}$$

where $\lambda_\pm = \sigma^2(1 \pm \sqrt{\gamma})^2$ are the bulk edges.

Table 1: Topological drift between early (2023) and late (2024) model cohorts. The bottleneck distance provides a formal lower bound on the Hausdorff distance between the capability point clouds via the stability theorem.

Metric	Value	Interpretation
Wasserstein-1 distance	**1.77**	Total structural change
Bottleneck distance	0.68	Maximum single-feature change
Hausdorff lower bound	≥ 0.34	Minimum geometric displacement
H_0 features (full cloud)	12	Connected components
H_1 features (full cloud)	0	No loops (small n)

Contamination detection. Eigenvalues exceeding λ_+ indicate structure beyond random variation. Under contamination, subjects with leaked data exhibit correlated performance boosts, creating additional outlier eigenvalues. We detect contamination via:

1. Compute eigenvalues of the sample covariance of the performance matrix.
2. Identify outliers exceeding Marchenko-Pastur bounds $[\lambda_-, \lambda_+]$.
3. Apply the Tracy-Widom test [**?**] to the largest eigenvalue for significance.
4. Localize contaminated subjects via leverage scores of outlier eigenvectors.

Proposition 1 (Contamination Signal)**.** *If k subjects are contaminated with correlated performance boost δ, the performance matrix gains an additional rank-k component, producing k eigenvalues exceeding λ_+ by $\Omega(n\delta^2)$.*

4 Experimental Results

We validate all five frameworks on MMLU capability data. All experiments use 12 models with performance profiles across 57 subjects, constructed from published benchmark results.

4.1 Framework 1: Topological Drift Detection

Persistence computation. We compute the Vietoris-Rips persistent homology of the full 12-model point cloud in $\mathbb{R}^{57}$. The persistence diagram reveals 12 H_0 features (connected components) with varying lifetimes and 0 H_1 features (loops), consistent with the small sample size relative to ambient dimension.

Temporal drift. We partition models into early (2023: GPT-4, CLAUDE-3 Opus, LLAMA-3 8B, PaLM-2, Mistral 7B, MIXTRAL) and late (2024: GPT-4o, Claude-3.5 Sonnet, LLAMA-3 70B, Qwen-2 72B, GEMINI 1.5 Pro, Phi-3 Medium) cohorts. table 1 reports the distances between persistence diagrams of the two cohorts.

The Wasserstein distance of 1.77 confirms genuine structural change between model generations. By Theorem 1, the Hausdorff distance between the point clouds is at least 0.34, establishing a formal lower bound on geometric displacement that no existing metric provides.

4.2 Framework 2: Sheaf-Theoretic Composition

Graph structure. Our evaluation sheaf is defined on a graph with 63 nodes (1 global, 5 categories, 57 subjects) and 65 edges. Categories comprise: STEM (17 subjects), Humanities (11), Social Sciences (11), Professional (10), Other (11).

Consistency analysis. table 2 reports consistency ratios for selected models. Higher ratios indicate more heterogeneous profiles where category-level averaging loses substantial information.

Key finding. GEMINI 1.5 Pro (ratio 1.83) and MIXTRAL (ratio 1.72) exhibit highly heterogeneous profiles where category-level averaging discards substantial information. For these models, reporting aggregate accuracy is formally misleading—the sheaf cohomology is non-trivial, indicating inconsistency between local and global evaluations. Conversely, LLAMA-3 8B (ratio 0.48) has a sufficiently uniform profile that aggregation is faithful.

Table 2: Sheaf consistency ratios for selected models. Higher values indicate that standard averaging loses more information about the model's true capability distribution. Models with heterogeneous profiles (ratio > 1) cannot be faithfully evaluated by aggregated scores.

Model	Consistency Ratio ρ	Profile Type
GEMINI 1.5 Pro	**1.83**	Highly heterogeneous
MIXTRAL	1.72	Heterogeneous
GPT-4o	1.15	Moderately heterogeneous
GPT-4	0.91	Moderate
CLAUDE-3 Opus	0.76	Relatively uniform
LLAMA-3 8B	0.48	Uniform

Table 3: Fisher-Rao distance statistics across all model pairs. The maximum distance (2.82) indicates substantial distributional difference, while geometrically close models with different raw accuracies reveal structure invisible to standard metrics.

Statistic	Value
Maximum pairwise distance	**2.82** (Mistral 7B $\leftrightarrow$ GPT-4o)
Mean pairwise distance	1.46
Minimum pairwise distance	0.31 (CLAUDE-3 Opus $\leftrightarrow$ GPT-4)

4.3 Framework 3: Information-Geometric Manifold

Fisher-Rao distances. We compute pairwise Fisher-Rao distances between all 12 models on the normalized capability simplex. table 3 reports summary statistics and notable findings.

Geometric structure. Multidimensional scaling (MDS) of the Fisher-Rao distance matrix reveals clear separation between capability tiers with interesting intermediate structure. Notably, CLAUDE-3 Opus and GPT-4 are geometrically close (distance 0.31) despite different raw accuracy values, indicating similar *distributional structure* across subjects. This finding is invisible to accuracy-based comparison and reveals shared architectural biases in capability allocation.

Fisher information traces. Weaker models (Mistral 7B, LLAMA-3 8B) exhibit higher Fisher information traces, indicating more peaked capability distributions concentrated on fewer subjects. These models are more sensitive to evaluation perturbation—a formal characterization of evaluation instability for weaker systems.

4.4 Framework 4: Failure Mode Landscape

Phase transition. We compute persistent homology of the failure correlation graph at increasing thresholds. table 4 shows the progression of topological features.

Interpretation. The emergence of H_1 features at $\tau = 0.8$ represents a genuine phase transition: below this threshold, failure patterns form a tree-like hierarchy (subjects cluster by difficulty), but above it, cyclic failure relationships appear. This indicates that at high difficulty thresholds, failure modes become structurally intertwined—models that fail on subject A tend to succeed on B, which tends to co-fail with C, which co-fails with A, forming non-trivial loops.

Failure correlation structure. The failure correlation matrix exhibits clear block structure aligned with subject categories, confirming that failures are structured (non-random) and revealing inter-category failure dependencies invisible to per-subject analysis.

4.5 Framework 5: Spectral Contamination Detection

Marchenko-Pastur analysis. For our performance matrix ($n = 12$ models, $d = 57$ subjects, $\gamma = d/n = 4.75$), the Marchenko-Pastur bounds are $[\lambda_- = 0.29, \lambda_+ = 2.13]$. table 5 summarizes the spectral analysis.

Table 4: Topological features of the failure landscape at increasing difficulty thresholds. A phase transition at $\tau = 0.8$ produces H_1 features (loops), indicating structurally distinct cyclic failure relationships that cannot be captured by hierarchical clustering.

Threshold τ	Failure Rate	β_0 (Components)	β_1 (Loops)	Interpretation
0.5	3.9%	1	0	Connected, no cycles
0.6	9.8%	1	0	Connected, no cycles
0.7	21.1%	1	0	Connected, no cycles
0.8	54.8%	1	**2**	Phase transition
0.9	82.4%	1	3	Rich cyclic structure

Table 5: Spectral analysis results for contamination detection. Clean data produces 4 outlier eigenvalues reflecting genuine capability structure. Synthetic contamination produces additional outliers detectable via Tracy-Widom statistics.

Metric	Clean Data	Contaminated Data
Marchenko-Pastur bounds $[\lambda_-, \lambda_+]$	[0.29, 2.13]	
Outlier eigenvalues ($> \lambda_+$)	4	7
Tracy-Widom statistic (largest λ)	**83.1**	127.4
p-value	$< 10^{-10}$	$< 10^{-10}$
Eigenvalue spacing	Between Wigner and Poisson	

Clean data analysis. The clean performance matrix exhibits 4 eigenvalues exceeding $\lambda_+ = 2.13$, reflecting genuine capability structure (model tiers, category-specific strengths). The Tracy-Widom statistic of 83.1 is highly significant ($p < 10^{-10}$), confirming real structure beyond random noise. The eigenvalue spacing lies between the Wigner surmise (correlated random matrix) and Poisson (independent entries), indicating moderate inter-task correlations.

Contamination injection. We inject synthetic contamination by boosting 5 randomly selected subjects by 15% for 4 models. The contaminated matrix produces 3 additional outlier eigenvalues, and the corresponding eigenvectors' leverage scores correctly identify the contaminated subjects. This demonstrates that spectral methods can detect contamination at a structural level, even when surface-level statistics remain plausible.

4.6 Comparison to Standard Metrics

table 6 summarizes the properties of our frameworks compared to standard evaluation approaches.

5 Discussion

5.1 Interpretation of Results

Our five frameworks demonstrate that mathematical tools from topology, sheaf theory, information geometry, and spectral analysis provide genuinely new information about LLM evaluation—information inaccessible to standard metrics.

Stability through topology. The topological drift detection framework provides something no existing evaluation metric offers: a formal guarantee that observed differences reflect genuine structural change rather than measurement noise. The stability theorem (Theorem 1) converts the Wasserstein distance of 1.77 into a provable lower bound on geometric displacement between model generations. This is qualitatively different from reporting that "accuracy increased by $X\%$"—it guarantees that the capability *landscape* has changed, not just a single summary statistic.

Composability through sheaves. The sheaf-theoretic framework formally characterizes a problem practitioners face intuitively: when can we trust aggregate scores? Our consistency ratio directly measures information loss under aggregation. GEMINI 1.5 Pro's ratio of 1.83 means that reporting its category-level scores discards 83% more information relative to signal than LLAMA-3 8B's

Table 6: Comparison of mathematical properties across evaluation approaches. Our frameworks are the first to provide formal stability guarantees, composability analysis, and spectral contamination detection for LLM evaluation.

Property	Accuracy	Elo Rating	TOPDRIFT	SHEAFEVAL	SPECCONTAM
Formal stability bound	×	×	✓	—	—
Composability guarantee	×	×	—	✓	—
Metric structure	×	partial	✓	—	✓
Contamination detection	×	×	—	—	✓
Multi-scale analysis	×	×	✓	✓	—
Distributional comparison	×	×	—	—	✓

uniform profile. This provides actionable guidance: models with high consistency ratios require fine-grained evaluation; aggregate leaderboard rankings are formally misleading for these systems.

Geometry beyond accuracy. The Fisher-Rao distance reveals model relationships invisible to accuracy comparison. The small distance (0.31) between CLAUDE-3 Opus and GPT-4, despite different overall accuracies, indicates shared capability *structure*—similar relative strengths and weaknesses across subjects. This suggests architectural or training data similarities that raw performance numbers cannot capture.

Phase transitions in failure. The emergence of H_1 features at threshold 0.8 reveals qualitative structure in the failure landscape. Below this threshold, failures decompose hierarchically; above it, cyclic dependencies emerge. This has practical implications: evaluation strategies that assume tree-structured failure modes (hierarchical decomposition, category-level analysis) break down for high-difficulty items.

Spectral fingerprints. The spectral framework demonstrates that contamination creates detectable structural signatures in the performance matrix eigenspectrum. Unlike surface-level methods that can be evaded by paraphrasing [**?**], spectral detection operates on the correlation structure of performance boosts, which is preserved even when individual items are modified.

5.2 Limitations

We identify several limitations that scope the applicability of our current results.

Simulated data. Model performance profiles are constructed from published accuracy scores rather than measured directly via API calls. While this provides realistic capability vectors, it does not capture the full distributional information (e.g., per-question correctness patterns) that would strengthen the topological and spectral analyses. Direct evaluation via model APIs is a necessary next step.

Small sample size. With 12 models in $\mathbb{R}^{57}$, the point cloud is sparse relative to ambient dimension. The absence of H_1 features in the full cloud reflects this limitation rather than the absence of genuine topological structure. Scaling to 100+ models using approximate persistence algorithms [**?**] would enable richer topological analysis.

Simplified sheaf construction. Our sheaf uses linear restriction maps (weighted averaging), which captures aggregation behavior but not the full complexity of evaluation composition. Nonlinear restriction maps and computation of higher sheaf cohomology would provide stronger guarantees but require more sophisticated algebraic machinery.

Contamination validation. We validate spectral detection using synthetic contamination injection rather than known real-world contamination cases. Validation against documented contamination events [**??**] would strengthen the practical applicability of this framework.

Computational cost. Persistent homology computation scales as $O(n^3)$ in the number of simplices, which grows exponentially with point cloud size. For production-scale deployment with thousands of models, approximate algorithms or dimensionality reduction would be necessary.

5.3 Broader Implications

Toward formal evaluation science. Our work suggests that LLM evaluation can and should be placed on formal mathematical foundations. Just as statistical testing provides guarantees for hypothesis evaluation, topological and geometric tools can provide guarantees for model evaluation. The stability theorem transforms evaluation from empirical measurement into provable characterization.

Complementary to existing approaches. We emphasize that our frameworks *complement* rather than replace existing evaluation methods. Standard accuracy metrics remain essential for absolute performance assessment. Our frameworks add a layer of structural analysis that detects when standard metrics are unreliable (high consistency ratios), when rankings are unstable (small persistence features), and when data integrity is compromised (spectral anomalies).

Evaluation as mathematical object. By treating evaluation data as mathematical objects (point clouds, sheaves, statistical manifolds, random matrices), we open the door to importing decades of theoretical results from pure mathematics. Each mathematical domain provides not just tools but *theorems*—formal guarantees that the evaluation community currently lacks.

6 Conclusion

We presented five mathematically grounded evaluation frameworks for large language models, drawing on persistent homology, sheaf theory, information geometry, topological data analysis, and random matrix theory. Through a comprehensive survey of 11 evaluation dimensions and proof-of-concept implementations on MMLU data, we demonstrated that these mathematical tools address fundamental gaps in current methodology: stability guarantees via the persistence stability theorem, composability analysis via sheaf cohomology, proper metric structure via Fisher-Rao geometry, structured failure detection via topological phase transitions, and contamination detection via spectral anomalies.

Our key finding is that mathematically rigorous evaluation is both feasible and informative. The Wasserstein distance of 1.77 between model generations, consistency ratios ranging from 0.48 to 1.83, Fisher-Rao distances up to 2.82, topological phase transitions at the 0.8 difficulty threshold, and Tracy-Widom statistics of 83.1 all provide quantitative insights inaccessible to standard accuracy metrics. These are not merely alternative numbers—they carry formal guarantees that standard metrics fundamentally cannot provide.

Future work. Three directions are immediate priorities. First, applying these frameworks to real model outputs via API evaluation would validate the approach on ground-truth data rather than published scores. Second, scaling topological analysis to 100+ models using approximate persistence algorithms would enable detection of richer topological features (H_1, H_2). Third, developing an open-source toolkit integrating all five frameworks would enable the evaluation community to adopt mathematically grounded methods without requiring expertise in algebraic topology or spectral theory. More broadly, we hope this work catalyzes a shift toward treating LLM evaluation as a formal mathematical discipline, with provable guarantees matching the rigor that the field's importance demands.

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